

# Beyond Naive INT8: Adaptive Post-Training Compression for 3D Gaussian Splatting

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## Abstract

3D Gaussian Splatting (3DGS) achieves high-quality real-time novel view synthesis but represents scenes as large sets of Gaussian primitives. Rather than proposing a new compression algorithm, this paper is a practical, diagnostic study of how post-training compression strategies behave inside the Nerfstudio/splatfacto ecosystem. We present an adaptive post-training optimization pipeline that compresses trained 3DGS models without retraining. The pipeline builds on a standard Nerfstudio splatfacto checkpoint and consists of four stages: adaptive opacity pruning using a per-scene 5th percentile sigmoid threshold, spatial and scale outlier clipping, spherical harmonic (SH) degree reduction with channel-blocked slicing, and quantization of the resulting Gaussian primitives. We evaluate our method across three distinct scenarios: sparse outdoor (Garden), dense outdoor (Bicycle), and indoor object capture (Vase). A mixed-precision configuration storing geometry at 16 bits and appearance at 8 bits reduces file size by 74% while incurring a minor Peak Signal-to-Noise Ratio (PSNR) drop of only 1.3 to 2.7 dB. We further diagnose the failure mode of naive 8-bit quantization, demonstrating that a single per-channel min/max range collapses geometry reconstruction quality to roughly 15 dB across all scenes and SH degrees because 256 levels cannot span the full scene extent. This degradation is a quantization-range artifact rather than a fundamental limit of 8-bit storage, as sub-group ranges recover quality to within 0.1 to 0.7 dB of FP16. This approach yields the best size and quality trade-off in this study, achieving a 78% reduction at a 1.7 dB cost on the Garden scene. The evaluation also yields a reproducible negative result: we document a failure mode of post-pruning fine-tuning in the Nerfstudio splatfacto trainer, tracing it to an optimizer-state dimension mismatch.

## CCS Concepts

• **Computing methodologies** → **Image-based rendering; Point-based models; Reconstruction.**

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## Keywords

3D Gaussian Splatting, novel view synthesis, model compression, pruning, quantization, spherical harmonics

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## 1 Introduction

3D Gaussian Splatting [1] has become a popular alternative to Neural Radiance Fields [2] for novel view synthesis. Instead of querying an implicit network per ray, 3DGS represents a scene as millions of anisotropic Gaussian primitives that can be rasterized in real time. The method delivers visual quality on par with Mip-NeRF 360 [3] but renders at orders of magnitude higher frame rates, rendering it highly suited for real-time applications in AR/VR, robotics, and free-viewpoint video.

The cost is storage. A typical outdoor scene trained with Nerfstudio's splatfacto implementation [5] contains 1–3 million Gaussians, each storing position, rotation, scale, opacity, and 16 view-dependent spherical-harmonic coefficients per color channel, yielding a single scene file of 300–700 MB.

Several recent works compress 3DGS through training-time changes. Among these, LightGaussian [6] introduces importance-aware pruning into the optimization loop, while Compact3DGS [7] learns a compact attribute representation. These approaches are effective but require modifying the training pipeline, which is a barrier for users who already have trained scenes or who use off-the-shelf trainers like Nerfstudio. A smaller body of work compresses scenes without retraining, where FlexGaussian [9] combines pruning with channel-wise mixed-precision quantization over many sub-group ranges, and Niedermayr et al. [8] apply sensitivity-aware vector quantization. Our work falls into this training-free category.

We take the post-training angle. We assume the user has an already trained checkpoint and ask: how much can we strip out, post hoc, without retraining? We propose a four-stage adaptive pipeline that operates entirely on the trained checkpoint. The pipeline is parameter light, runs in under a minute on commodity hardware, and produces a compressed PLY file that loads in standard viewers. Where our quantization design aligns with the sub-group scheme of FlexGaussian [9], we explicitly document this connection.

**Positioning, and who this is for.** This paper does not propose a fundamentally new compression algorithm; the operators we compose—opacity pruning, outlier clipping, SH degree reduction,

and mixed-precision or sub-group quantization—are each grounded in prior art. We contribute a systematic account of how they behave when applied post hoc inside a widely used trainer ecosystem, together with two findings visible only from that vantage point: a mechanistic diagnosis of why the obvious naive 8-bit implementation collapses, and a reproducible negative result showing that the natural remedy, fine-tuning the pruned model, silently does nothing in the current splatfacto trainer. The gap addressed is a deployment one: 3DGS quality is already adequate for many applications, but hundreds of megabytes per scene is prohibitive for web viewers, mobile clients, and large scene archives, and prior work largely resolves this during training, which does not help users who already hold finished checkpoints. Our audience is therefore practitioners operating trained splatfacto scenes—AR/VR and web-delivery developers, robotics and digital-heritage teams, and researchers maintaining scene libraries—who need a drop-in post-processing step requiring no retraining, no learned parameters, and no change to the training code; Section 5.8 states which variant suits which delivery target.

Our contributions are:

- An adaptive opacity pruning rule based on the 5th percentile of the sigmoid-transformed opacity distribution, which produces scene-dependent thresholds without manual tuning.
- A channel-blocked SH slicing scheme that reduces the SH degree while preserving the layout constraints of PLY storage formats.
- A diagnosis of why naive 8-bit quantization of 3DGS geometry fails: a single per-channel min/max range cannot represent positions spanning the full scene extent at 8-bit granularity, collapsing quality to roughly 15 dB uniformly across scenes and SH degrees. We show this is a range artifact, not a limit of 8-bit storage, by recovering quality to within 0.1–0.7 dB of FP16 with sub-group ranges, confirming that the sub-group quantization used by training-free methods such as FlexGaussian [9] transfers to post-training operation in the Nerfstudio/splatfacto ecosystem.
- A reproducible negative result: zero measurable benefit from post-pruning fine-tuning in the Nerfstudio splatfacto trainer, traced to optimizer state dimension mismatch.
- Validation on three scenes covering distinct capture conditions, with full ablations across SH degree and quantization combinations.

To support reproducibility, we will release our full pipeline implementation and evaluation configurations publicly upon acceptance.

## 2 Related Work

**Neural radiance fields.** NeRF [2] reconstructs scene geometry and appearance by optimizing an MLP to map view rays to radiance and density values. Later work improved quality [3] and convergence speed [4], but rendering throughput remained orders of magnitude below rasterization.

**3D Gaussian Splatting.** Kerbl et al. [1] replaced the implicit MLP with an explicit set of anisotropic 3D Gaussians, each with view-dependent color encoded as spherical harmonics. Training

**Table 1: Scope comparison. Reported reduction is on Mip-NeRF 360 under each method’s own protocol and is *not* directly comparable across rows. Our distinct elements are the failure diagnosis and the fine-tuning no-op result on splatfacto.**

| Method            | Training-free | Targets splatfacto | Diagnoses INT8 fail | Reports FT no-op | Reported reduction |
|-------------------|---------------|--------------------|---------------------|------------------|--------------------|
| LightGaussian [6] | ✗             | ✗                  | ✗                   | ✗                | ~15×               |
| Compact3DGS [7]   | ✗             | ✗                  | ✗                   | ✗                | ~25×               |
| FlexGaussian [9]  | ✓             | ✗                  | ✗                   | ✗                | up to 96.4%        |
| Ours              | ✓             | ✓                  | ✓                   | ✓                | 74–91%             |

optimizes Gaussian parameters by differentiable rasterization. Nerfstudio’s splatfacto [5] provides an open implementation that we build upon.

**Pruning.** LightGaussian [6] introduces an importance-based pruning mechanism into the optimization loop to reduce primitive counts, and Compact3DGS [7] employs learnable masks during training. Unlike these techniques, which require training-time modifications, our pruning routine is entirely post-processing, operating strictly on the final checkpoint by utilizing raw opacity statistics.

**Quantization and compression.** Several methods reduce per Gaussian storage. Niedermayr et al. [8] apply sensitivity-aware vector quantization. FlexGaussian [9] is a training-free method that pairs pruning with channel-wise mixed-precision quantization, dividing each attribute channel into many sub-group ranges so that 8-bit quantization remains accurate. Our quantization stage shares this sub-group insight. Rather than presenting it as novel, we use it to explain a failure that a naive single range implementation produces: we show that per-channel global min/max quantization of geometry collapses quality to roughly 15 dB, characterize the mechanism, and demonstrate that sub-group ranges recover it. This diagnostic framing, together with confirmation that the fix operates post-training within Nerfstudio’s splatfacto, is our quantization contribution.

**Spherical harmonic reduction.** The contribution of higher-order SH coefficients to perceptual quality is uneven and often small. Several systems support degree reduction at export. We characterize the quality cost of reducing degree from 3 to 1 across our three test scenes and identify per-scene sensitivity.

## 2.1 Positioning

We do not provide head-to-head quantitative comparisons against LightGaussian [6], Compact3DGS [7], or FlexGaussian [9], because their published results were produced under different training pipelines, resolutions, and evaluation splits than our post-hoc splatfacto setting; we instead compare scope (Table 1) and report each method’s reduction as the authors stated it.

## 3 Method

### 3.1 Pipeline overview

We assume access to a trained Nerfstudio splatfacto checkpoint at the end of 30,000 iterations. The checkpoint stores Gaussian parameters as flat tensors under `_model.gauss_params`: means

( $n \times 3$ ), `features_dc` ( $n \times 3$ ), `features_rest` ( $n \times 15 \times 3$ ), `opacities` ( $n \times 1$ , in logit space), `scales` ( $n \times 3$ , in log space), and `quats` ( $n \times 4$ ).

Our pipeline applies four post-training stages, each operating on the output of the previous one: adaptive opacity pruning, spatial and scale clipping, SH degree reduction, and quantization (evaluated under several schemes, Section 3.5). The result is written as a standard PLY file for use with existing 3DGS viewers, and is also injected back into a copy of the checkpoint so that we can run Nerfstudio’s evaluator on the compressed model for fair PSNR/SSIM/LPIPS comparison.

### 3.2 Adaptive opacity pruning

The opacities stored in the checkpoint are in logit space. To prune Gaussians whose contribution to rendering is negligible, we transform to alpha space with the sigmoid function and discard primitives below a per-scene threshold  $\tau$ .

The training-time threshold used by splatfacto’s `cull-alpha-thresh` is a fixed 0.005, applied during densification. We found this value too conservative for post-training pruning: it leaves many low-impact Gaussians in scenes with denser primitive distributions, while being roughly correct for sparse scenes. Instead, we set  $\tau$  adaptively as the 5th percentile of the sigmoid transformed opacity distribution:

$$\tau = \text{percentile}_5(\sigma(\mathbf{o})), \quad (1)$$

where  $\mathbf{o}$  is the vector of stored logit opacities and  $\sigma(\cdot)$  is the sigmoid function. This rule trims the lowest contributing 5% of Gaussians regardless of the underlying distribution shape, which naturally adapts to scene density.

### 3.3 Spatial and scale outlier clipping

Trained scenes often contain floating outlier primitives that have no impact on visible rendering but expand the axis-aligned bounding boxes utilized by interactive viewers, leading to increased memory footprints. We mitigate this by clipping Gaussians with centers situated outside the 0.5 to 99.5 percentile interval along each spatial coordinate axis.

A second class of outliers is anisotropic Gaussians with extreme scale. In log space these can reach values that, after exponentiation, produce primitives several orders of magnitude larger than the scene. We clip stored log-scales at the 1–99 percentile interval and apply an absolute upper bound of  $\log 1.0 = 0$  as a defensive cap. In the Vase scene the absolute cap did not trigger; in the Bicycle scene it removed several pathological splats responsible for visible smearing.

### 3.4 SH degree reduction with channel-blocked slicing

The `features_rest` tensor stores 15 SH coefficients per color channel for degree 3. Reducing to degree 2 requires 8 coefficients per channel; degree 1 requires 3; degree 0 keeps only the DC term.

The PLY format stores these as a flat array `f_rest_0` through `f_rest_44`, laid out as 15 R-channel coefficients followed by 15 G-channel coefficients followed by 15 B-channel coefficients. Naively keeping the first  $3k$  scalars (for target degree giving  $k$  coefficients per channel) does not retain the first  $k$  coefficients of each channel

and produces a corrupted file. We instead slice per channel:

$$\text{keep} = \{15c + i \mid c \in \{0, 1, 2\}, i \in \{0, \dots, k - 1\}\}. \quad (2)$$

When loading a reduced degree PLY back into the checkpoint for evaluation, we zero pad `features_rest` to  $(n, 15, 3)$  so that the model accepts the tensor.

### 3.5 Quantization

We analyze three quantization schemes applied to the pruned, SH-reduced primitives; the ablation tables additionally include uniform FP16 and uniform INT8 as reference points.

**Mixed precision (FP16 geometry, INT8 appearance).** Geometry parameters means, scales, quaternions, and opacities are highly sensitive to precision because small numerical errors translate directly into visible misalignment of splats. Appearance parameters DC color and higher-order SH coefficients are far more tolerant. We therefore store geometry in 16-bit floating point and appearance in 8-bit fixed point with per-channel min/max scaling.

**Per-column INT8 (baseline, fails).** A naive 8-bit scheme stores every field, including geometry, in 8-bit fixed point using a single minimum and maximum per channel. As Section 5.6 shows, this collapses reconstruction quality. For a geometry channel whose values span the full scene extent (on the order of meters), 256 quantization levels correspond to positional steps of several centimeters, which is far coarser than the sub-pixel accuracy splats require.

**Sub-group INT8 (fixes it).** The remedy is to give each channel many local ranges instead of one global range. We sort primitives along a Morton (Z-order) curve so that array-adjacent Gaussians are spatially adjacent, partition each channel into  $G \approx 1000$  contiguous sub-groups, and quantize each sub-group with its own min/max. Storage stays 8-bit; only the per-group range metadata is added, which is negligible relative to the payload. This is the same sub-group principle used by FlexGaussian [9]; we apply it post-training and contrast it directly against the per-column scheme on identical inputs.

For appearance fields, the min and max values are written into the PLY header as comments so that a reader can dequantize on load:

```
comment quant_minmax f_dc_0 -2.137 1.984
```

The PLY format does not natively support 16-bit floating point. We work around this by interpreting the 16-bit pattern as `uint16`:

```
fp16 = orig.astype(np.float16)
out_arr[fname] = fp16.view(np.uint16)
```

This yields a file of identical size to true FP16. For loading back into the model for evaluation, the values are roundtripped through `float16` so that the dequantized representation matches what a viewer would see. We note that SuperSplat [10] cannot render these files because it interprets the `uint16` fields as integers; quantized files are intended for downstream readers that respect the header comments. The unquantized variants remain SuperSplat compatible.

## 4 Experimental Setup

### 4.1 Datasets and scenes

We evaluate on three scenes representing distinct capture conditions:

- **Garden** from the Mip-NeRF 360 dataset [3]: outdoor, sparse, plant-heavy.
- **Bicycle** from the Mip-NeRF 360 dataset: outdoor, dense, with fine bicycle spokes that stress reconstruction.
- **Vase**: a self-captured indoor scene of a single object, 79 frames from an iPhone X subsampled from a 237-frame orbital video, processed with COLMAP exhaustive matching.

The Vase scene tests whether the pipeline generalizes beyond the Mip-NeRF 360 outdoor captures to a different setting (indoor, single object, phone-captured); we treat it as a generalization probe, not a contribution in itself.

**Rationale for these data sources and this protocol.** Our study is computational, so the analogue of data collection is the choice of scenes, training protocol, and measurement procedure. Garden and Bicycle come from Mip-NeRF 360 [3], the benchmark under which the methods we position against [6, 7, 9] report results, keeping our observations legible against the literature. Alternatives such as Tanks and Temples or Deep Blending were considered but share its professionally captured, well-registered character, adding scenes without adding a capture condition. Because the failure modes we study are properties of the trained Gaussian distribution, and a consumer-grade capture yields visibly different opacity and scale distributions, we captured Vase ourselves as first-hand primary data; no existing dataset supplies that condition. We report PSNR, SSIM, and LPIPS rather than one metric because they are complementary—PSNR is sensitive to the geometric misplacement quantization induces, LPIPS to the perceptual smearing of Figure 4—and are standard for novel view synthesis. Metrics come from Nerfstudio’s own ns-eval on held-out views, and every variant within a scene derives from one checkpoint, isolating compression effects from training variance (Section 4.2).

### 4.2 Training and infrastructure

All training was done with Nerfstudio’s splatfacto model for 30,000 iterations on a single NVIDIA T4 GPU (15 GB VRAM) hosted on Kaggle. We use `cull-alpha-thresh 0.005`, `max-gauss-ratio 5.0`, and `downscale-factor 4` on the COLMAP-processed input. Images at native resolution are hidden during training to prevent automatic re-downscaling.

### 4.3 Evaluation

We report PSNR, SSIM, LPIPS, file size in megabytes, and rendered framerate (FPS) on the held-out evaluation split. PSNR, SSIM, and LPIPS are computed with Nerfstudio’s ns-eval command after the compressedPLY has been brain-swapped back into a copy of the original checkpoint. FPS is measured at the same resolution used for evaluation.

**Table 2: Garden scene (1,572,747 baseline Gaussians,  $\tau = 0.0138$ ). Sub-group INT8 recovers the per-column INT8 collapse to near-FP16 quality at the smallest file size.**

| Variant                  | Size (MB) | PSNR  | SSIM  | LPIPS | FPS  |
|--------------------------|-----------|-------|-------|-------|------|
| baseline                 | 390.0     | 27.56 | 0.855 | 0.096 | 14.3 |
| pruned_sh3               | 347.9     | 26.29 | 0.835 | 0.117 | 17.0 |
| pruned_sh3_fp16          | 173.9     | 26.24 | 0.832 | 0.119 | 17.2 |
| pruned_sh3_int8          | 87.0      | 14.92 | 0.257 | 0.632 | 16.9 |
| pruned_sh3_int8_subgroup | 85.7      | 25.82 | 0.810 | 0.126 | 17.2 |
| pruned_sh3_mixed         | 101.0     | 26.23 | 0.832 | 0.119 | 17.0 |
| pruned_sh2               | 230.1     | 25.34 | 0.817 | 0.136 | 17.1 |
| pruned_sh2_fp16          | 115.0     | 25.30 | 0.814 | 0.137 | 16.8 |
| pruned_sh2_int8          | 57.5      | 14.94 | 0.262 | 0.632 | 16.7 |
| pruned_sh2_int8_subgroup | 55.2      | 25.01 | 0.794 | 0.144 | 17.2 |
| pruned_sh2_mixed         | 71.5      | 25.30 | 0.814 | 0.138 | 16.8 |
| pruned_sh1               | 145.9     | 24.25 | 0.792 | 0.158 | 17.2 |
| pruned_sh1_fp16          | 72.9      | 24.23 | 0.790 | 0.159 | 16.8 |
| pruned_sh1_int8          | 36.5      | 14.93 | 0.264 | 0.633 | 16.6 |
| pruned_sh1_int8_subgroup | 33.4      | 24.02 | 0.772 | 0.164 | 16.8 |
| pruned_sh1_mixed         | 50.5      | 24.23 | 0.790 | 0.159 | 16.7 |

**Table 3: Bicycle scene (2,649,045 baseline Gaussians,  $\tau = 0.0105$ ). The dense outdoor capture incurs a larger pruning cost than the other scenes but tolerates SH reduction well; sub-group INT8 again recovers the per-column collapse.**

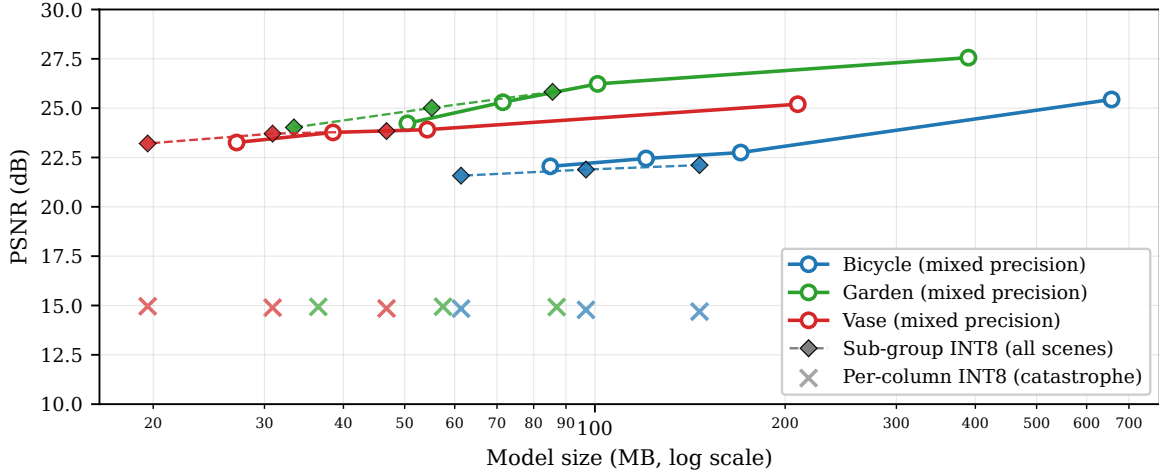
| Variant                  | Size (MB) | PSNR  | SSIM  | LPIPS | FPS  |
|--------------------------|-----------|-------|-------|-------|------|
| baseline                 | 657.0     | 25.44 | 0.771 | 0.160 | 13.2 |
| pruned_sh3               | 585.7     | 22.77 | 0.726 | 0.200 | 15.2 |
| pruned_sh3_fp16          | 292.9     | 22.75 | 0.724 | 0.201 | 15.3 |
| pruned_sh3_int8          | 146.4     | 14.69 | 0.251 | 0.642 | 15.2 |
| pruned_sh3_int8_subgroup | 146.4     | 22.11 | 0.680 | 0.216 | 15.4 |
| pruned_sh3_mixed         | 170.1     | 22.75 | 0.724 | 0.202 | 15.3 |
| pruned_sh2               | 387.3     | 22.47 | 0.713 | 0.214 | 15.3 |
| pruned_sh2_fp16          | 193.7     | 22.45 | 0.711 | 0.215 | 15.1 |
| pruned_sh2_int8          | 96.8      | 14.78 | 0.256 | 0.642 | 15.2 |
| pruned_sh2_int8_subgroup | 96.8      | 21.89 | 0.669 | 0.229 | 15.3 |
| pruned_sh2_mixed         | 120.5     | 22.45 | 0.710 | 0.215 | 15.0 |
| pruned_sh1               | 245.6     | 22.07 | 0.691 | 0.231 | 15.2 |
| pruned_sh1_fp16          | 122.8     | 22.06 | 0.690 | 0.232 | 15.2 |
| pruned_sh1_int8          | 61.4      | 14.84 | 0.260 | 0.643 | 15.2 |
| pruned_sh1_int8_subgroup | 61.4      | 21.58 | 0.651 | 0.245 | 15.4 |
| pruned_sh1_mixed         | 85.0      | 22.05 | 0.689 | 0.232 | 15.1 |

## 5 Results

### 5.1 Per-scene ablations

Tables 2, 3, and 4 report the full ablation for each scene. “pruned\_shN” denotes the output of the pruning, clipping, and SH-reduction stages at target degree  $N$  in FP32. “\_fp16” applies uniform 16-bit floating-point quantization. “\_int8” applies uniform 8-bit fixed-point quantization to all fields. “\_mixed” applies our mixed-precision scheme (FP16 geometry, INT8 appearance).





**Figure 1: Cross-scene Pareto frontier of post-training compression; solid lines connect Pareto-optimal points per scene. Per-column INT8 variants are excluded as catastrophic outliers (Section 5.6); sub-group INT8 lies on or near the frontier at the smallest sizes for all three scenes.**

**Table 4: Vase scene (844,544 baseline Gaussians,  $\tau = 0.0148$ ). As with the other scenes, per-column INT8 collapses and sub-group INT8 recovers to near-FP16 quality at the smallest file size.**

| Variant                  | Size (MB) | PSNR  | SSIM  | LPIPS | FPS  |
|--------------------------|-----------|-------|-------|-------|------|
| baseline                 | 209.4     | 25.20 | 0.814 | 0.126 | 18.0 |
| pruned_sh3               | 187.0     | 23.93 | 0.794 | 0.155 | 19.9 |
| pruned_sh3_fp16          | 93.5      | 23.92 | 0.793 | 0.155 | 19.2 |
| pruned_sh3_int8          | 46.8      | 14.85 | 0.387 | 0.492 | 19.0 |
| pruned_sh3_int8_subgroup | 46.8      | 23.84 | 0.787 | 0.159 | 19.1 |
| pruned_sh3_mixed         | 54.3      | 23.91 | 0.792 | 0.156 | 18.7 |
| pruned_sh2               | 123.7     | 23.78 | 0.782 | 0.167 | 19.5 |
| pruned_sh2_fp16          | 61.8      | 23.77 | 0.781 | 0.167 | 18.7 |
| pruned_sh2_int8          | 30.9      | 14.89 | 0.395 | 0.491 | 19.7 |
| pruned_sh2_int8_subgroup | 30.9      | 23.71 | 0.776 | 0.171 | 19.7 |
| pruned_sh2_mixed         | 38.5      | 23.77 | 0.781 | 0.168 | 19.0 |
| pruned_sh1               | 78.4      | 23.27 | 0.763 | 0.183 | 19.7 |
| pruned_sh1_fp16          | 39.2      | 23.26 | 0.762 | 0.183 | 18.7 |
| pruned_sh1_int8          | 19.6      | 14.96 | 0.399 | 0.493 | 19.8 |
| pruned_sh1_int8_subgroup | 19.6      | 23.21 | 0.757 | 0.186 | 19.9 |
| pruned_sh1_mixed         | 27.1      | 23.26 | 0.762 | 0.183 | 19.8 |

## 5.2 Headline results

The mixed-precision configuration `pruned_sh3_mixed` gives a consistent operating point across scenes: 74.1% reduction on Garden (390.0 to 101.0 MB,  $-1.33$  dB), 74.1% on Bicycle (657.0 to 170.1 MB,  $-2.69$  dB), and 74.1% on Vase (209.4 to 54.3 MB,  $-1.29$  dB). The quality cost is scene-dependent—the dense Bicycle capture loses more to pruning than the sparser scenes—but the size reduction is stable near 74%. Figure 2 shows how the file size changes through successive pipeline stages for each scene.

The best size/quality point overall is sub-group INT8. On Garden, `pruned_sh3_int8_subgroup` reaches 85.7 MB (78.0% reduction)

at 25.82 dB, a cost of only 1.74 dB and smaller than the mixed variant. Pushing to `pruned_sh1_int8_subgroup` reaches 33.4 MB (91.4% reduction) at 24.02 dB. Bicycle follows the same pattern at 146.4 MB / 22.11 dB (77.7% reduction). A more conservative 16-bit configuration, `pruned_sh3_fp16`, gives 55.4% reduction on both outdoor scenes (Garden  $-1.32$  dB, Bicycle  $-2.69$  dB; Figure 1).

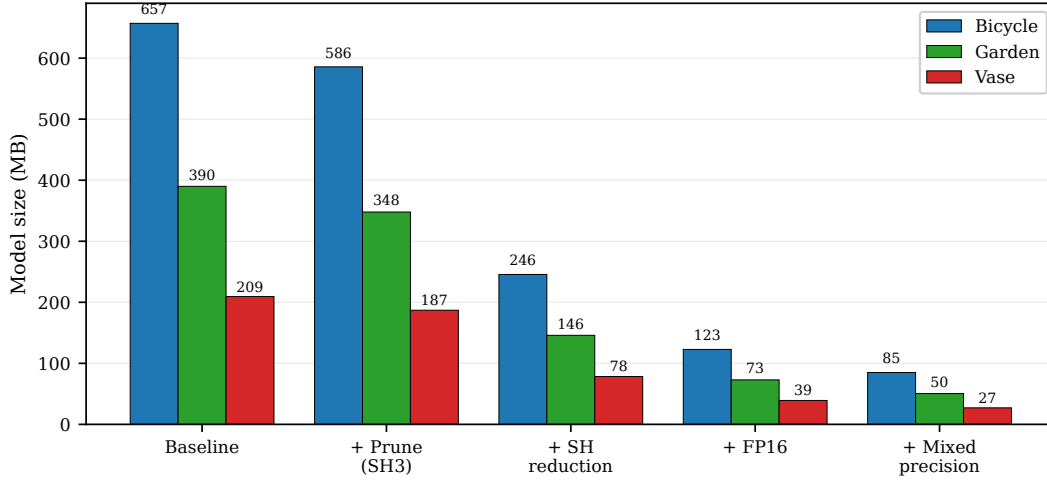
## 5.3 Cross-scene adaptive threshold

The adaptive opacity threshold produces a distinct value for each scene, which supports the case that a single hand-tuned threshold does not generalize. Values are: 0.0105 for Bicycle (dense outdoor), 0.0138 for Garden (sparse outdoor), and 0.0148 for Vase (indoor object) (Figure 3). For reference, `splatfacto`’s training-time `cull-alpha-thresh` default is 0.005, which is well below all three. A fixed post-training threshold of 0.005 would under-prune all three scenes.

## 5.4 Sensitivity to the pruning percentile

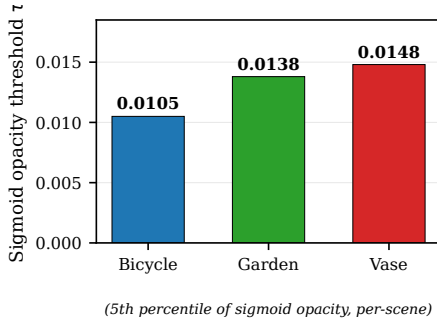
The 5th-percentile rule of Section 3.2 fixes the pipeline’s one free parameter, so we measure how strongly results depend on it. Holding every other stage fixed (clipping, SH degree 3, identical protocol, all variants derived from the original trained checkpoints), we sweep  $p \in \{1, 2.5, 5, 7.5, 10\}$  on Garden and Bicycle (Table 5). The sweep is restricted to these two scenes because consistency with Tables 2–4 requires the original checkpoint, which was not retained for Vase; a re-trained substitute would not be comparable.

The pipeline is insensitive to this choice. Across a tenfold range of  $p$ , PSNR varies by 0.26 dB on Garden and 0.36 dB on Bicycle, declining smoothly alongside size: each successive step costs at most 0.09 dB on Garden and 0.14 dB on Bicycle. No threshold produces sharp degradation, so the adopted  $p=5$  lies in a broad flat region rather than at a tuned optimum, and practitioners can move within this range and predict the cost approximately linearly. The reported results therefore do not rest on a sensitive hyperparameter—the



Mixed precision shown as final stage for all three scenes; sub-group INT8 (not shown) is smaller still on Garden and Bicycle.

Figure 2: Model size through the pipeline stages, ending at the SH1 mixed-precision variant; sub-group INT8 (not plotted) is smaller still on Garden and Bicycle.



(5th percentile of sigmoid opacity, per-scene)

Figure 3: Per-scene adaptive opacity threshold (5th percentile of  $\sigma(\text{opacity})$  from the trained model). Dense outdoor captures select a more conservative threshold than sparse outdoor and indoor object scenes, reflecting the wider opacity distribution in scenes with high splat density.

property that matters for a method meant to run unattended on unseen checkpoints—while the rule still yields distinct per-scene thresholds at every  $p \geq 2.5$ .

### 5.5 Mixed precision approaches FP16 quality

Across every SH degree and all three scenes, `_mixed` and `_fp16` variants produce PSNR, SSIM, and LPIPS that agree to within 0.01 dB and 0.001 respectively (Tables 2–4). The mixed variant is smaller because the appearance fields, which dominate file size, are stored at 8 bits while geometry stays at 16. This indicates that 8-bit quantization of SH coefficients is effectively lossless and that it is the precision of geometry, not appearance, that governs reconstruction quality—the observation that motivates the analysis in the next subsection.

Table 5: Sensitivity to the pruning percentile  $p$  (SH3, other stages fixed;  $p=5$  adopted). Quality and size vary smoothly, with no sharp degradation threshold.

| $p$  | $\tau$ | Garden    |       |        | Bicycle   |       |  |
|------|--------|-----------|-------|--------|-----------|-------|--|
|      |        | Size (MB) | PSNR  |        | Size (MB) | PSNR  |  |
| 1.0  | 0.0041 | 362.7     | 26.38 | 0.0040 | 610.8     | 22.85 |  |
| 2.5  | 0.0089 | 357.2     | 26.35 | 0.0047 | 601.6     | 22.86 |  |
| 5.0  | 0.0138 | 347.9     | 26.29 | 0.0105 | 585.7     | 22.77 |  |
| 7.5  | 0.0195 | 338.7     | 26.21 | 0.0159 | 569.8     | 22.63 |  |
| 10.0 | 0.0275 | 329.5     | 26.12 | 0.0225 | 554.1     | 22.50 |  |

### 5.6 Per-column INT8 fails; sub-group INT8 recovers it

Every per-column `_int8` variant collapses to a PSNR of roughly 14–15 dB, with SSIM falling to 0.25–0.40 and LPIPS rising to 0.49–0.64, on all three scenes and at all SH degrees (Tables 2–4). The uniformity of the collapse—PSNR varies by less than 0.2 dB across SH degrees within a scene—is itself diagnostic: the loss is independent of appearance content, which points to geometry rather than color as the cause. Quantizing means, scales, and quaternions with a single per-channel min/max range maps the full scene extent onto 256 levels. For a positional channel spanning meters, each level is several centimeters, far coarser than the sub-pixel placement splats require, and the result is the severe smearing visible in Figure 4.

This is a property of the *range*, not of 8-bit storage. Replacing the single global range with  $G \approx 1000$  Morton-ordered sub-group ranges (Section 3.5) keeps storage at 8 bits but lets each local region use its own scale. The effect is large and consistent across all three scenes: at SH degree 3, sub-group INT8 recovers Garden from 14.92 to 25.82 dB (+10.9 dB), Bicycle from 14.69 to 22.11 dB (+7.4 dB), and Vase from 14.85 to 23.84 dB (+9.0 dB), in every case landing within

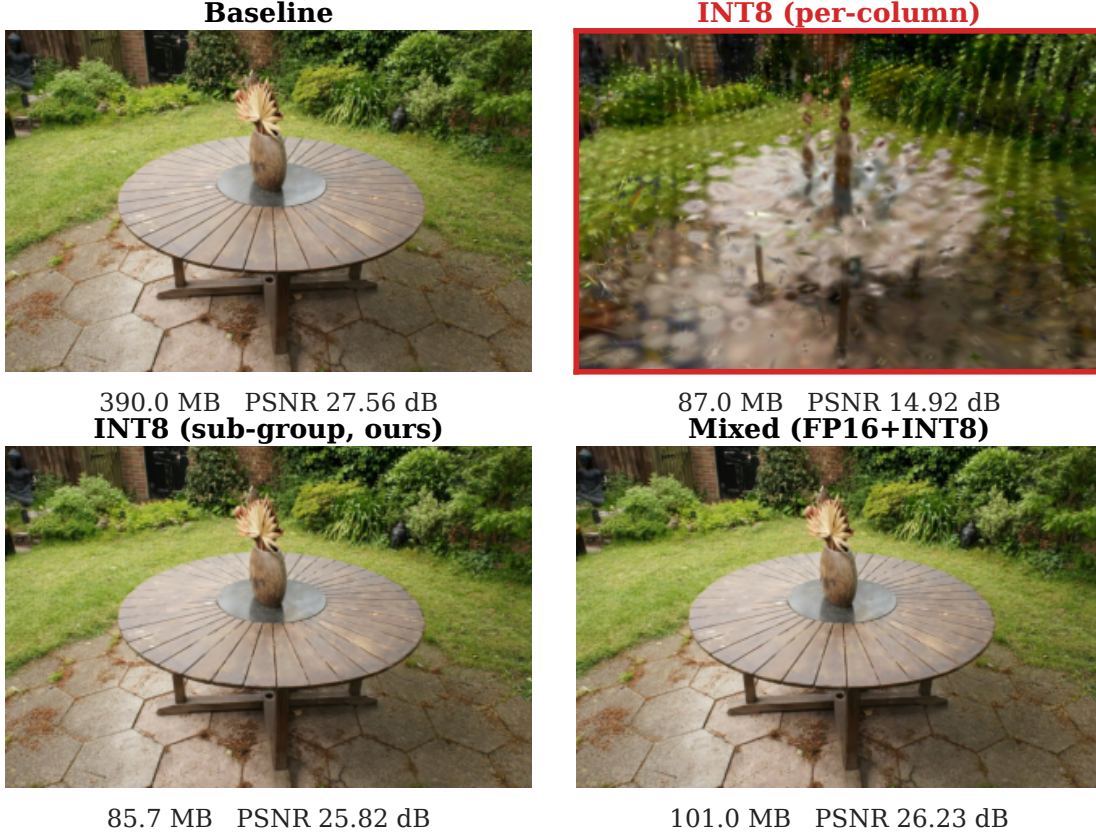
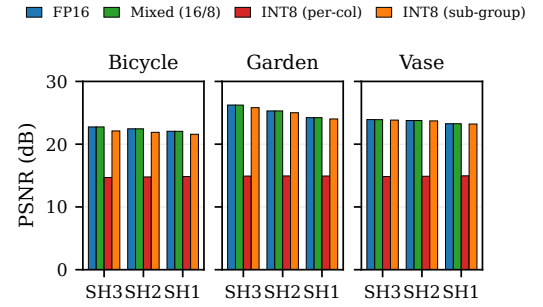


Figure 4: Garden, identical eval view, four storage schemes. Per-column INT8 (top right, red) collapses geometry into smearing at 14.92 dB, while sub-group INT8 (bottom left) is visually indistinguishable from the baseline while occupying essentially the same file size as the per-column variant (85.7 vs 87.0 MB), recovering 25.82 dB. Mixed precision is shown for reference.

0.1–0.7 dB of the FP16 result at the same 8-bit budget. The recovery holds across all SH degrees (Figure 5). This confirms that the sub-group quantization principle used by FlexGaussian [9] transfers to a post-training setting in the Nerfstudio/splatfacto ecosystem, and it reframes the apparent “INT8 catastrophe” as an implementation artifact of naive range selection rather than a fundamental limit.

### 5.7 Post-pruning fine-tuning produces no improvement

We attempted to recover the PSNR lost to pruning by resuming splatfacto training on the pruned checkpoint for an additional 2,000–10,000 iterations across four configurations of learning rate and densification settings. In all four runs, parameter updates after resume were below numerical noise and final-eval PSNR was unchanged from the pre-finetune value. The cause appears to be optimizer state dimension mismatch: the saved optimizer state references the original tensor sizes and silently no-ops on the resized tensors. A correct implementation would require rebuilding the optimizer state, which the splatfacto trainer does not support cleanly. We report this as a negative result; practitioners seeking to recover pruned quality by fine-tuning should consider it incompatible with the current Nerfstudio splatfacto trainer.



Per-column INT8 collapses (~15 dB); sub-group recovers to near-FP16.

Figure 5: Reconstruction quality by quantization scheme and SH degree, for all three scenes. Per-column INT8 (red) collapses uniformly to roughly 15 dB regardless of SH degree, while sub-group INT8 (orange) recovers to within a fraction of a dB of FP16 and mixed precision in every case.

### 5.8 Discussion

Our findings carry several practical consequences:

**SH reduction cost is scene dependent and often small.** The PSNR cost of reducing SH degree from 3 to 2 is 0.15 dB on Vase, 0.30 dB on Bicycle, and 0.95 dB on Garden; reducing from 3 to 1 costs 0.66, 0.70, and 2.04 dB respectively. View-dependence varies with content, so SH degree should be chosen per scene rather than fixed. Reducing SH3→2 is nearly free only for the matte indoor object; outdoor scenes pay a measurable but modest cost.

**Geometry precision is what matters for quantization.** 8-bit appearance storage is effectively lossless, but 8-bit geometry is unusable under a single global range and only becomes usable with sub-group ranges or 16-bit storage. The geometry/appearance asymmetry, not the bit width alone, is the governing factor.

**Adaptive thresholding generalizes.** A single percentile rule produces appropriate, distinct thresholds across scene types without per-scene tuning.

**Choosing a configuration.** *FP16* is the conservative option: no measurable quality cost beyond pruning, half the file, and—since the attribute layout is unchanged—the safest choice when the target viewer is unknown or the scene must stay editable in existing tools, suiting archival storage and iteration. *Mixed precision* is our recommended delivery default: it matches *FP16* to within 0.01 dB at a consistent 74% reduction while requiring the reader to dequantize only appearance fields, the simplest form of reader support. *Sub-group INT8* is for when size dominates—bandwidth-constrained streaming, mobile clients, large libraries—reaching the smallest sizes here at 0.1–0.7 dB relative to *FP16*, but requiring a reader that honors the per-group range metadata, so it fits pipelines under the user’s own control.

**Storage reduction does not translate into proportional speedup.** A 74–91% smaller file might be expected to render faster, yet measured FPS moves only slightly (Tables 2–4). This is expected, and stating it delimits what the pipeline is for. Rasterization cost is governed by how many primitives must be sorted and blended and how many pixels each covers, not by on-disk compactness: our quantization stages change storage format only, and every attribute is dequantized to working precision at load, so an *INT8* and an *FP32* scene present the rasterizer identical work. Only pruning affects runtime, and it is deliberately conservative, leaving most of the rasterization load intact; the roughly 1–3 FPS we do observe follows from the reduced primitive count and smaller bounding volume. The pipeline targets storage, bandwidth, and load time—the binding constraints for delivery—rather than steady-state throughput.

## 5.9 Limitations

Our evaluation covers three scenes; a controlled, shared-protocol comparison against training-time methods (LightGaussian, Compact3DGS) and the training-free FlexGaussian is the main step needed to position this pipeline against the state of the art, and is left for future work. Several further factors bound the scope of our claims. Because all scenes are trained and evaluated at downscale-factor 4, our absolute PSNR values are not comparable to full-resolution 3DGS results and should be read only relative to our own per-scene baseline; the trends we report—the per-column collapse, the sub-group recovery, and the geometry/appearance asymmetry—are what transfer, not the headline numbers. The sub-group count  $G \approx 1000$  is fixed rather than tuned per scene, so the recovered

sub-group quality is a conservative lower bound rather than an optimum. Three scenes is a small sample; although they span sparse outdoor, dense outdoor, and indoor object capture, broader validation across the full Mip-NeRF 360 suite and other datasets remains future work. Finally, our *FP16* storage reinterprets the 16-bit pattern as *uint16* in the *PLY*, which is correct for readers that respect the header comments but is not understood by SuperSplat: the unquantized variants render correctly there, while the quantized *PLY*s require a compatible reader.

## 6 Conclusion

We presented a post-training pipeline that compresses 3DGS models by 74–91%, at a PSNR cost of 1.3–2.7 dB for the recommended mixed-precision configuration, with no retraining and under a minute of runtime. Our central finding is diagnostic: the apparent failure of 8-bit quantization is an artifact of a single per-channel range spanning the whole scene, and sub-group ranges recover quality to within 0.1–0.7 dB of *FP16* at the smallest sizes in our study. We further show the result is insensitive to the pipeline’s one free parameter, and report a negative result—zero benefit from post-pruning fine-tuning in *splatfacto*, traced to optimizer-state dimension mismatch. The pipeline emits standard *PLY* files and is intended as a drop-in tool for users who already have trained scenes.

## 7 Reproducibility

All experiments use Nerfstudio’s *splatfacto* trained for 30,000 iterations on a single NVIDIA T4. All fifteen variants per scene derive from one trained checkpoint, so within-scene ablations are mutually consistent by construction; metrics are computed with *ns-eval* after each compressed *PLY* is injected back into a copy of that checkpoint. The pipeline uses no learned parameters and runs in under a minute on commodity hardware. The full implementation, per-scene result files, and figure scripts will be released publicly upon acceptance.

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